

Network Flow II

(ADA 2023, CSE 222)

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April 6, 2023

Outline

- ① Ford-Fulkerson's Algorithm
- ② Application to Bipartite Matching

- **Flow Network:** A directed graph $G = (V, E)$ with $s, t \in V(G)$ (s is source and t is sink). Every edge $e \in E(G)$ has a nonnegative *capacity* $c : E(G) \rightarrow \mathbb{Z}^+$.
- **Defining flow:** What is *flow* or traffic in a network? An s - t flow in a network is a function $f : E(G) \rightarrow \mathbb{Z}^+$ satisfying the following two conditions.
 - **Capacity condition:** For every $e \in E(G)$, $f(e) \leq c(e)$.
 - **Flow conservation condition:** For each u other than s and t , we have

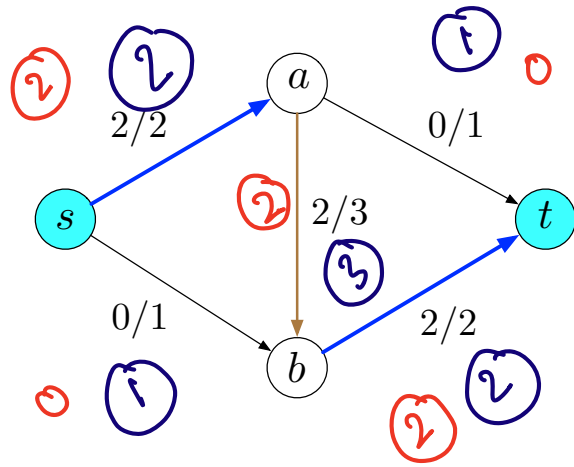
$$\sum_{v:(v,u) \in E(G)} f(v, u) = \sum_{v:(u,v) \in E(G)} f(u, v)$$

More specifically $f^{\text{out}}(u) = \sum_{v:(u,v) \in E(G)} f(u, v) = \sum_{e \text{ out of } u} f(e)$ and

$$f^{\text{in}}(u) = \sum_{v:(v,u) \in E(G)} f(v, u) = \sum_{e \text{ into } u} f(e).$$

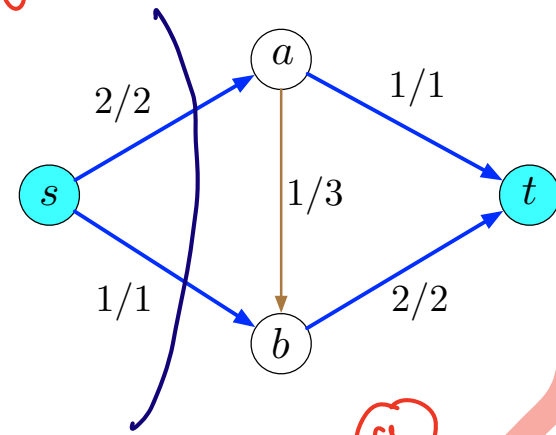
- **Input:** A flow network with capacity conditions flow conservation conditions.
- **Output:** A flow $f : E(G) \rightarrow \mathbb{Z}^+$ that satisfies capacity conditions and flow conservation conditions, and maximize $\sum_{e \text{ out of } s} f(e) = \sum_{u:(s,u) \in E(G)} f(s, u)$.

Examples of Flow



(S, T)

$s \in S$
 $t \in T$



$S = \{s, a\}$
 $T = \{b, t\}$

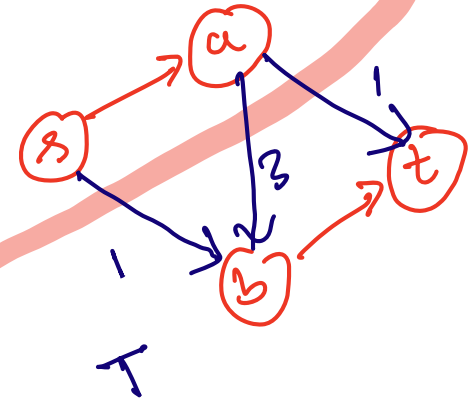
$c(S, T)$

$S = \{s, b\}$
 $T = \{a, t\}$

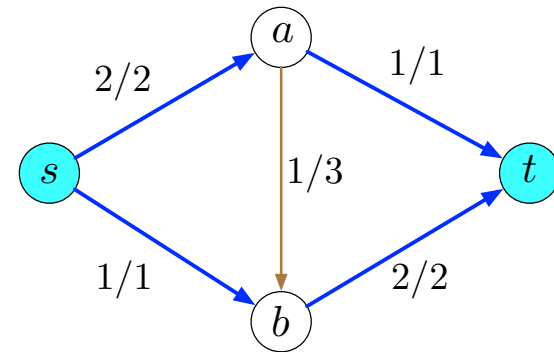
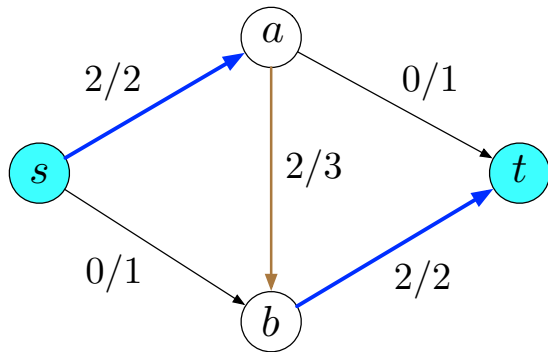
$c(u, v)$

$\sum_{u \in S} \sum_{v \in T}$

$S = \{s\}$
 $T = \{a, b, t\}$

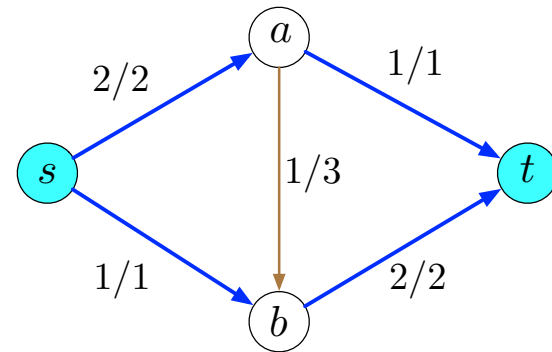
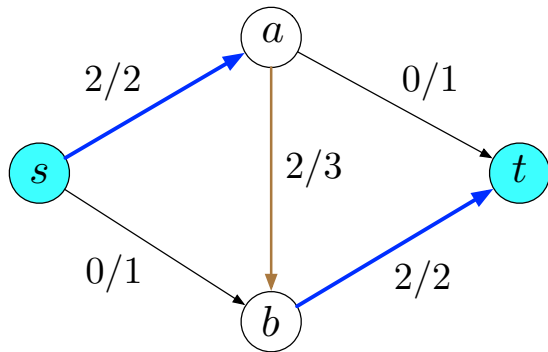


Examples of Flow



- For a valid (s, t) -flow f ,
$$\sum_{u:(s,u) \in E(G)} f(s, u) = \sum_{u:(u,t) \in E(G)} f(u, t).$$

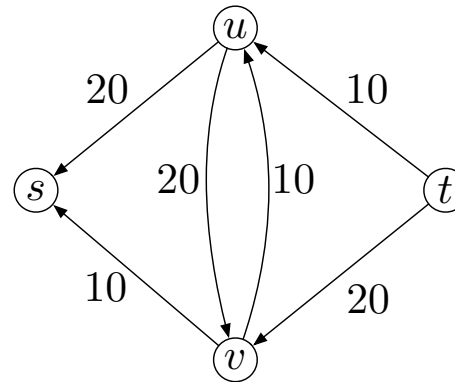
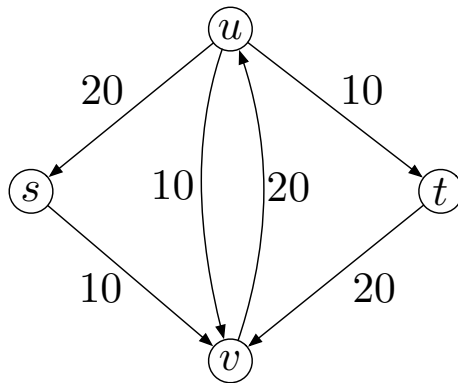
Examples of Flow



- For a valid (s, t) -flow f , $\sum_{u:(s,u) \in E(G)} f(s, u) = \sum_{u:(u,t) \in E(G)} f(u, t)$.
- $$\sum_{u:(s,u) \in E(G)} f(s, u) = \sum_{u:(s,u) \in E(G)} f(s, u) - \sum_{v \in V \setminus \{s,t\}} \left(\sum_{(u,v) \in E(G)} f(u, v) - \sum_{(v,w) \in E(G)} f(v, w) \right) = \sum_{u:(u,t) \in E(G)} f(u, t)$$

Total flow coming out of s is equal to the total flow going into t .

- Let G be a flow network and a f be a flow in G . Then, G_f has same vertex set as G .
- For every edge e , if $0 < f(e) < c(e)$, then $f(e) - c(e)$ is called *residual capacity*. By this forward edge e , this residual amount of flow $c(e) - f(e)$ can still be sent.
- For every edge e , if $0 < f(e) < c(e)$, then a backward edge e^{rev} is created with $c(e^{rev}) = f(e)$ in G_f .



For all $e \in E(G)$, initially $f(e) \leftarrow 0$ and $G_f = G$;

while there is an s - t path P in G_f **do**

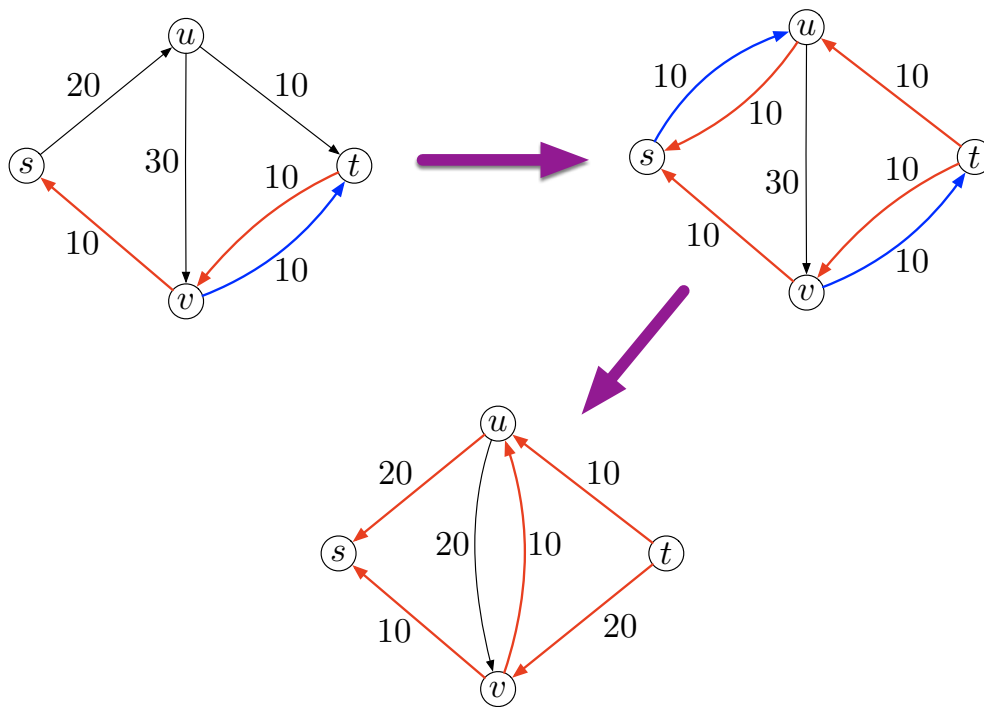
$b \leftarrow \min\{c(e) \mid e \in P\}$;

 Increase the flow of b at every edge in P ;

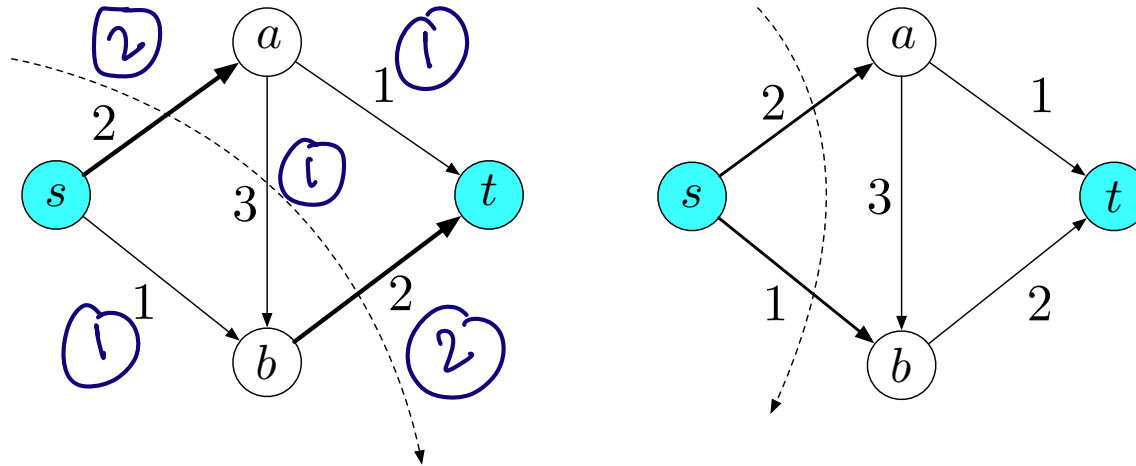
 Construct G_f accordingly;

end

Return f ;



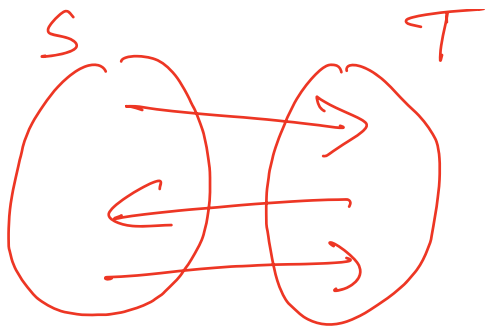
Cut in Digraphs



- In a flow network (G, s, t, c) , an (s, t) -cut is a nonempty $S \subset V(G)$ such that $s \in S$, and $t \in V \setminus S$.
- Capacity of an (s, t) -cut (S, T) is $\sum_{u \in S} \sum_{v \in T} c(u, v)$.
- Capacity of $(\{s, b\}, \{a, t\})$ is 4 and capacity of $(\{s\}, \{a, b, t\})$ is 3.

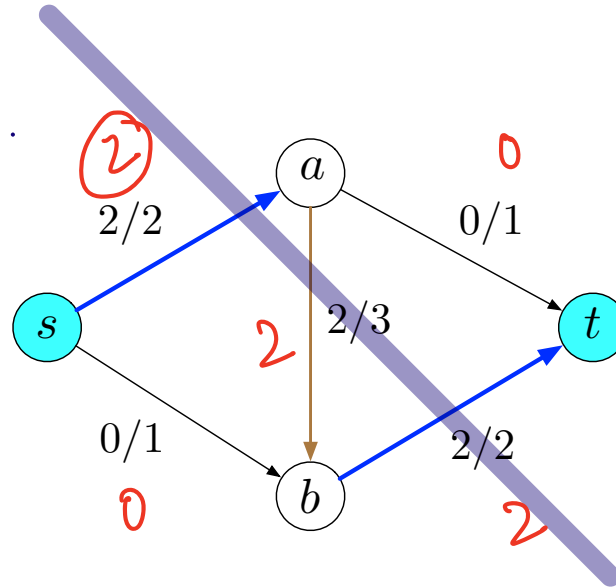
$$S = \{s\}, T = \{a, b, t\}$$

$$f(S, T) = \sum_{u \in S} \sum_{v \in T} f(u, v)$$



$$- \sum_{u \in T} \sum_{v \in S} f(u, v)$$

Flow across a cut



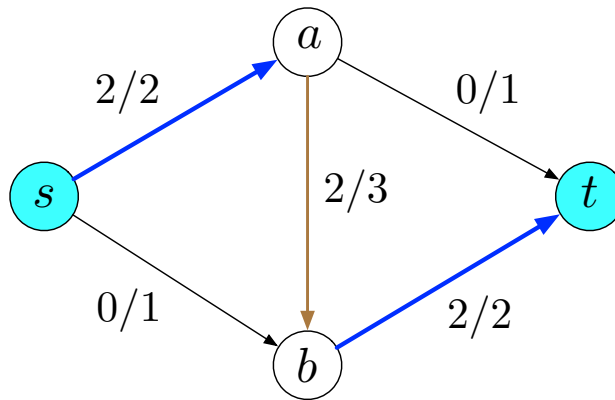
- $f(S, T) = \sum_{u \in S} \sum_{v \in T} f(u, v) - \sum_{u \in S} \sum_{v \in T} f(v, u)$ [flow across (S, T) -cut].

$$S = \{s\}, T = \{a, b, t\} \quad f(S, T) = 2$$

$$S = \{s, a\}, T = \{b, t\} \quad f(S, T) = 2$$

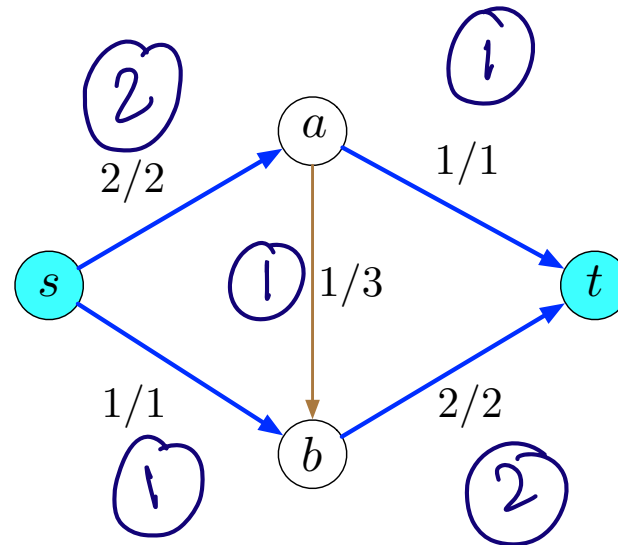
$$S = \{s, b\}, T = \{a, t\} \quad = 2$$

Flow across a cut



- $f(S, T) = \sum_{u \in S} \sum_{v \in T} f(u, v) - \sum_{u \in S} \sum_{v \in T} f(v, u)$ [flow across (S, T) -cut].
- Flow across $(\{s, b\}, \{a, t\})$ -cut is **2**.

Flow across a cut



- $f(S, T) = \sum_{u \in S} \sum_{v \in T} f(u, v) - \sum_{v \in T} \sum_{u \in S} f(v, u)$ [flow across (S, T) -cut].
- Flow across $(\{s\}, \{a, b, t\})$ -cut is **3**.
- Flow across $(\{s, a\}, \{b, t\})$ -cut is also **3**.

$$c(S, T) = 4$$

$$f(S, T) = 3$$

$$S = \{s, b\} \quad T = \{a, t\}$$

$$S = \{s\} \quad T = \{a, b, t\}$$

$$f(S, T) = 3 = c(S, T)$$

Max-Flow Min-Cut Theorem

Let (G, s, t, c) be a flow network and $f: E(G) \rightarrow \mathbb{Z}^+$ be a valid (s, t) -flow. Then, the followings are equivalent.

- ① $\text{val}(f)$ is a maximized.
- ② G_f has no augmenting paths.
- ③ There exists an (s, t) -cut (S, T) such that $c(S, T) = \text{val}(f)$.

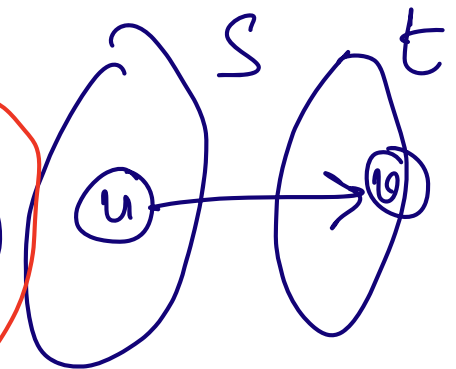
$$\text{val}(f) = \sum_{u: (s, u) \in E} f(s, u)$$

First we prove

- Let f be a flow and (S, T) be an (s, t) -cut in G . Then, $f(S, T) \leq c(S, T)$, and

$$f(S, T) = \sum_{u \in S} \sum_{v \in T} f(u, v)$$

$$- \sum_{v \in T} \sum_{u \in S} f(v, u)$$



$$\leq \sum_{u \in S} \sum_{v \in T} f(u, v)$$

$$\leq \sum_{u \in S} \sum_{v \in T} c(u, v)$$

$$= c(S, T)$$

Proof Sketch

Let f be a flow and (S, T) be an (s, t) -cut in G . Then, $f(S, T) \leq c(S, T)$.

$$f(S, T) = \sum_{u \in S} \sum_{v \in T} f(u, v) - \sum_{v \in T} \sum_{u \in S} f(v, u)$$

Proof Sketch

Let $\mathbf{f}$ be a flow and (S, T) be an (s, t) -cut in G . Then, $\mathbf{f}(S, T) \leq \mathbf{c}(S, T)$.

$$\begin{aligned}\mathbf{f}(S, T) &= \sum_{u \in S} \sum_{v \in T} \mathbf{f}(u, v) - \sum_{v \in T} \sum_{u \in S} \mathbf{f}(v, u) \\ &\leq \sum_{u \in S} \sum_{v \in T} \mathbf{f}(u, v) \leq \sum_{u \in S} \sum_{v \in T} \mathbf{c}(u, v)\end{aligned}$$

Proof Sketch

Let f be a flow and (S, T) be an (s, t) -cut in G . Then, $f(S, T) \leq c(S, T)$.

$$\begin{aligned} f(S, T) &= \sum_{u \in S} \sum_{v \in T} f(u, v) - \sum_{v \in T} \sum_{u \in S} f(v, u) \\ &\leq \sum_{u \in S} \sum_{v \in T} f(u, v) \leq \sum_{u \in S} \sum_{v \in T} c(u, v) \\ &= c(S, T) \end{aligned}$$

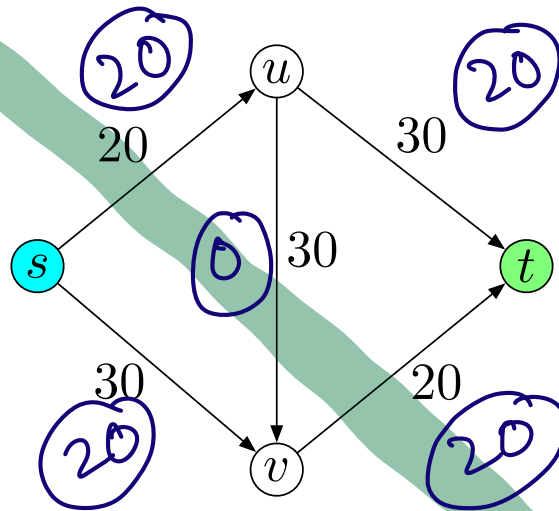
Max-Flow Min-Cut Theorem

$$S = \{s\}$$

$$T = \{a, b, t\}$$

$$f(S, T)$$

$$c(S, T)$$



$$S = \{s, u\}$$

$$T = \{u, t\}$$

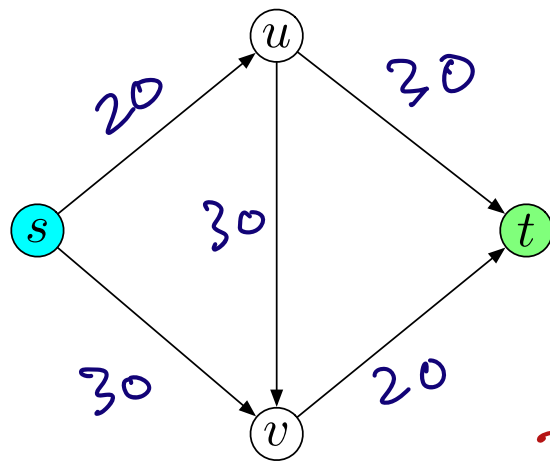
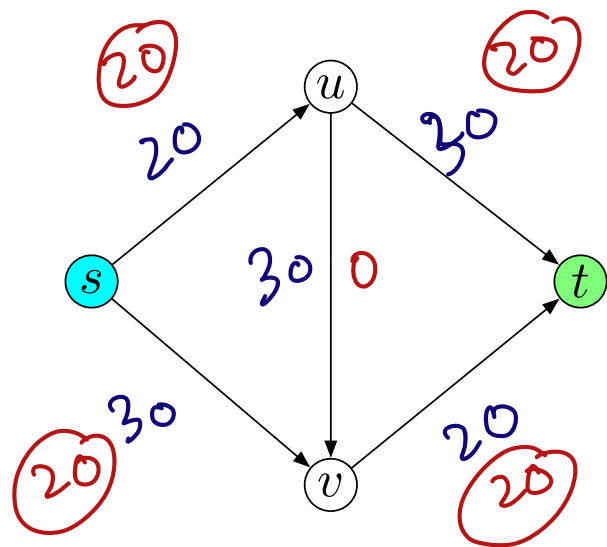
$$f(S, T) = 40$$

$$c(S, T) = 40$$

Let (G, s, t, c) be a flow network and $f: E(G) \rightarrow \mathbb{Z}^+$ be a valid (s, t) -flow. Then, the followings are equivalent.

- ① $\text{val}(f)$ is a maximized.
- ② G_f has no augmenting paths.
- ③ There exists an (s, t) -cut (S, T) such that $c(S, T) = \text{val}(f)$.

$$f(S, T) = f(s, u) + f(u, t) - f(u, v)$$



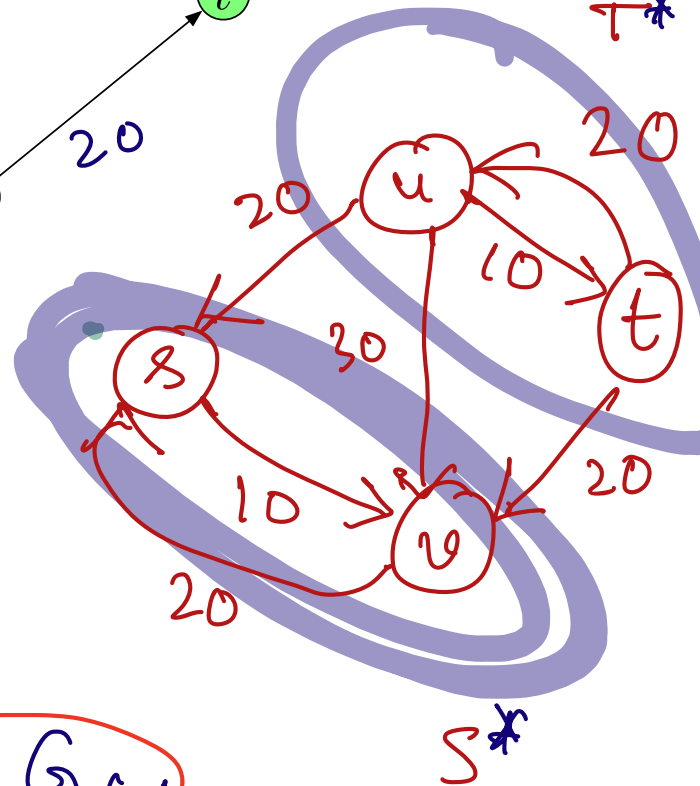
$$S^* = \{s, v\}$$

$$T^* = \{u, t\}$$

Construct an (s, t) -cut (S^*, T^*)
as follows.

S^* = the set of vertices that
are reachable from s in G_f .

$T^* = V(G) \setminus S^*$. clearly, $t \notin S^*$ because?
 f is a maximum flow.



$$f(S^*, T^*) = \sum_{u \in S^*} \sum_{v \in T^*} f(u, v)$$

$$= c(S^*, T^*) \quad \leftarrow \sum_{u \in S^*} \sum_{v \in T^*} c(u, v)$$

